Fakulteta za informacijske študije
Faculty of information studies

lab05

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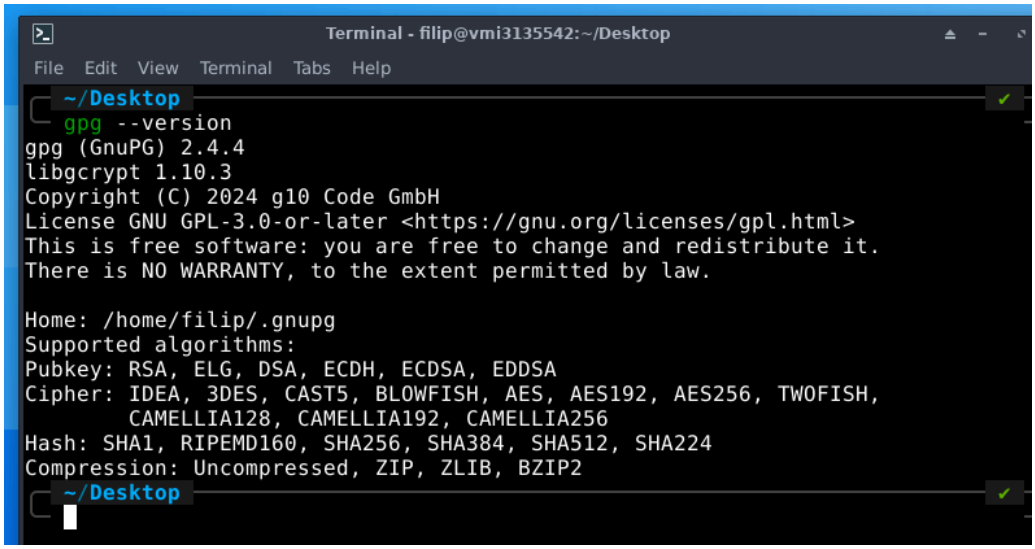

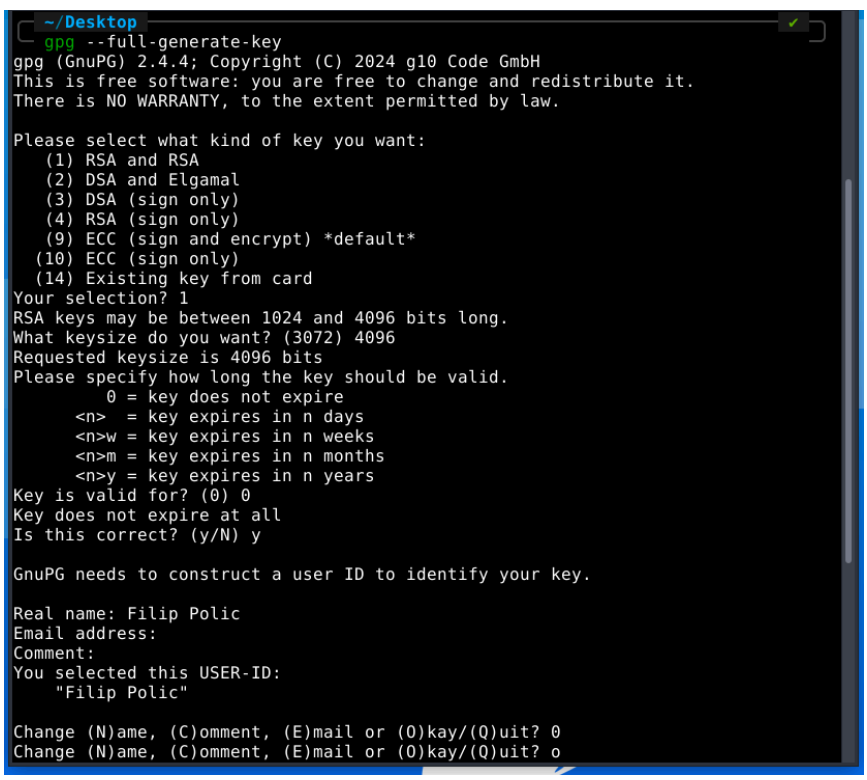

```
pub   rsa4096 2026-04-27 [SC]
      C6E20C681C39F87C3D044D100DF165F6A3DF57E2
uid           [ultimate] Alice <alice@example.com>
sub   rsa4096 2026-04-27 [E]

pub   rsa4096 2026-04-27 [SC]
      4312266E790235F8EC12B64FAF202C5881303B7E
uid           [ultimate] Bob <bob@example.com>
sub   rsa4096 2026-04-27 [E]

pub   rsa4096 2026-04-27 [SC]
      D6FB9A633CCCF2813A8F0B0EFF86BF3F6A4BA1C8
uid           [ultimate] Mallory <mallory@example.com>
sub   rsa4096 2026-04-27 [E]

~/Desktop
```

```
~/Desktop
gpg --fingerprint alice@example.com
pub   rsa4096 2026-04-27 [SC]
      C6E2 0C68 1C39 F87C 3D04  4D10 0DF1 65F6 A3DF 57E2
uid           [ultimate] Alice <alice@example.com>
sub   rsa4096 2026-04-27 [E]

~/Desktop
gpg --fingerprint bob@example.com
pub   rsa4096 2026-04-27 [SC]
      4312 266E 7902 35F8 EC12  B64F AF20 2C58 8130 3B7E
uid           [ultimate] Bob <bob@example.com>
sub   rsa4096 2026-04-27 [E]

~/Desktop
gpg --fingerprint mallory@example.com
pub   rsa4096 2026-04-27 [SC]
      D6FB 9A63 3CCC F281 3A8F  0B0E FF86 BF3F 6A4B A1C8
uid           [ultimate] Mallory <mallory@example.com>
sub   rsa4096 2026-04-27 [E]

~/Desktop
```

```
~/Desktop                                                    ✔ | 8s
gpg --armor --export mallory@example.com > bob_pubkey.asc
~/Desktop                                                    ✔
gpg --import bob_pubkey.asc
gpg: key FF86BF3F6A4BA1C8: "Mallory <mallory@example.com>" not changed
gpg: Total number processed: 1
gpg:              unchanged: 1
~/Desktop                                                    ✔
```

```
~/Desktop                                                    ✔ | 8s
gpg --armor --export mallory@example.com > bob_pubkey.asc
~/Desktop                                                    ✔
gpg --import bob_pubkey.asc
gpg: key FF86BF3F6A4BA1C8: "Mallory <mallory@example.com>" not changed
gpg: Total number processed: 1
gpg:              unchanged: 1
~/Desktop                                                    ✔
echo "Confidential message for Bob" > secret.txt
~/Desktop                                                    ✔
gpg --encrypt --recipient mallory@example.com secret.txt
~/Desktop                                                    ✔
gpg --decrypt secret.txt.gpg
gpg: encrypted with rsa4096 key, ID 20DB145B8BCEFB47, created 2026-04-27
      "Mallory <mallory@example.com>"
Confidential message for Bob
~/Desktop                                                    ✔ | 6s
```

```
~/Desktop                                                    ✔ | 6s
gpg --fingerprint mallory@example.com
pub   rsa4096 2026-04-27 [SC]
      D6FB 9A63 3CCC F281 3A8F  0B0E FF86 BF3F 6A4B A1C8
uid           [ultimate] Mallory <mallory@example.com>
sub   rsa4096 2026-04-27 [E]

~/Desktop                                                    ✔
gpg --fingerprint bob@example.com
pub   rsa4096 2026-04-27 [SC]
      4312 266E 7902 35F8 EC12  B64F AF20 2C58 8130 3B7E
uid           [ultimate] Bob <bob@example.com>
sub   rsa4096 2026-04-27 [E]

~/Desktop                                                    ✔
```

```
gpg> trust
sec  rsa4096/AF202C5881303B7E
     created: 2026-04-27  expires: never       usage: SC
     trust: ultimate      validity: ultimate
ssb  rsa4096/048FA5A2F69ED29E
     created: 2026-04-27  expires: never       usage: E
[ultimate] (1). Bob <bob@example.com>

Please decide how far you trust this user to correctly verify other users' keys
(by looking at passports, checking fingerprints from different sources, etc.)

  1 = I don't know or won't say
  2 = I do NOT trust
  3 = I trust marginally
  4 = I trust fully
  5 = I trust ultimately
  m = back to the main menu

Your decision? 5
Do you really want to set this key to ultimate trust? (y/N) y

sec  rsa4096/AF202C5881303B7E
     created: 2026-04-27  expires: never       usage: SC
     trust: ultimate      validity: ultimate
ssb  rsa4096/048FA5A2F69ED29E
     created: 2026-04-27  expires: never       usage: E
[ultimate] (1). Bob <bob@example.com>

gpg> quit
~/Desktop                                                    ✔ | 44s
```

```
~/Desktop                                                    ✔ | 4s
gpg --check-sigs bob@example.com
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid:   6  signed:   0  trust: 0-, 0q, 0n, 0m, 0f, 6u
gpg: next trustdb check due at 2027-04-22
pub   rsa4096 2026-04-27 [SC]
      4312266E790235F8EC12B64FAF202C5881303B7E
uid           [ultimate] Bob <bob@example.com>
sig!3        AF202C5881303B7E 2026-04-27  [self-signature]
sig!         BEC972E47E3C105B 2026-04-27  peer@example.com
sub   rsa4096 2026-04-27 [E]
sig!         AF202C5881303B7E 2026-04-27  [self-signature]

gpg: 3 good signatures
~/Desktop                                                    ✔
gpg --check-sigs bob@example.com
pub   rsa4096 2026-04-27 [SC]
      4312266E790235F8EC12B64FAF202C5881303B7E
uid           [ultimate] Bob <bob@example.com>
sig!3        AF202C5881303B7E 2026-04-27  [self-signature]
sig!         BEC972E47E3C105B 2026-04-27  peer@example.com
sub   rsa4096 2026-04-27 [E]
sig!         AF202C5881303B7E 2026-04-27  [self-signature]

gpg: 3 good signatures
~/Desktop                                                    ✔
```

Why doesn't GPG detect MITM attacks automatically?

GPG не може сам да зна да ли јавни кључ стварно припада Bob-у. Он само види да постоји јавни кључ са одређеним идентитетом. Ако нападач пошаље свој кључ уместо Bob-овог, GPG то не може аутоматски препознати без провере fingerprint-а или поверења у кључ.

What is a fingerprint and why is it important?

Fingerprint је јединствени отисак јавног кључа. Важан је јер омогућава кориснику да провери да ли је јавни кључ заиста исправан и да ли припада правој особи.

Why is email not a secure channel for exchanging keys?

Email није увек сигуран јер порука може бити пресретнута или измењена. Нападач може послати свој јавни кључ и представити га као Bob-ов. Зато је боље fingerprint проверити другим каналом, на пример уживо или телефоном.

How does the Web of Trust reduce the risk of MITM attacks?

Web of Trust смањује ризик тако што корисници потписују кључеве других корисника након провере fingerprint-а. Ако је кључ потписан од стране особе којој верујемо, већа је шанса да је тај кључ исправан.